1 An A4-sized paper is measured using a meter scale. Which of the following represents the most precise measurement?

A. 0.3 m
B. 30 cm
C. 29.71 cm
D. 29.7 cm

2 Two spherical hailstones with radii in the ratio $\frac{1}{2}$ fall freely from a great height through the atmosphere. Then their momentum in the ratio after they attain terminal velocity is _____

A. $\frac{1}{2}$
B. $\frac{1}{4}$
C. $\frac{1}{16}$
D. $\frac{1}{32}$

3 The difference between the maximum and minimum kinetic energy in a vertical circular motion is _____

A. 5mgr
B. 2mgr
C. 3mgr
D. mgr

4 The nuclear density (ρ) varies with the atomic mass number (A) as _____

A. $\rho \propto A^0$
B. $\rho \propto A^1$
C. $\rho \propto A^2$
D. $\rho \propto A^3$

5 Two bodies of different masses possess equal kinetic energy. Then their linear momentum will be _____

A. Equal
B. Greater for the heavier body
C. Greater for the lighter body
D. Depends on the direction of motion

6 The resistance of a wire is R. If the length of the wire is halved and its diameter is doubled, what will be the new resistance?

A. $\frac{R}{4}$
B. $\frac{R}{2}$
C. $\frac{R}{8}$
D. $\frac{R}{16}$

7 An electron is moving towards the North and experiences a force in the vertically upward direction. The direction of the magnetic field at that point is in the direction of _____

A. East
B. West
C. Upward
D. North

8 A diffraction pattern is obtained using a beam of red light. If the red light is replaced by blue light, then _____

A. Diffraction band becomes narrower and get crowded together
B. Bands become broader and farther apart
C. Bands disappear
D. No effect

9 If a galvanometer is shunted by $(\frac{1}{n})^{th}$ of the value of resistance, then the fraction of the total current passing through galvanometer is _____

A. $\frac{1}{n+1}$
B. $\frac{n}{n+1}$
C. 1
D. $\frac{1}{n}$

10 An observer is moving towards a stationary sound source. The observer perceives a sound of higher frequency. This happens because _____

A. Relative velocity of sound increases
B. Relative velocity of sound decreases
C. Wavelength of sound increases
D. Wavelength of sound decreases

11 During full wave rectification, the current in each diode flows until _____

A. half cycle of input signal
B. complete full cycle of input signal
C. more than half cycle of input signal
D. one-fourth of input signal

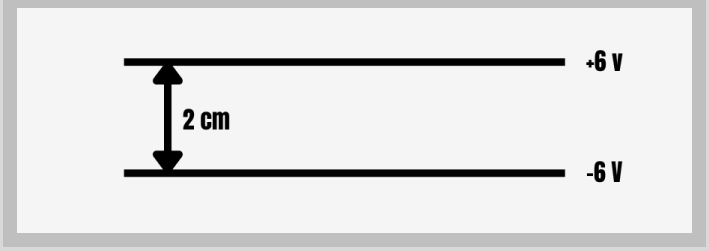
12 An unpolarized light is moving along positive Y-axis, then the direction of electric field vector is along _____

A. +ve X-direction
B. +ve Z-direction
C. -ve X-direction
D. Either X-direction or Z-direction

13 In Nepal, the AC mains supply has an RMS voltage of 220V. What is the peak (maximum) voltage?

A. $220 \times \sqrt{2}$
B. $\frac{220}{\sqrt{2}}$
C. 220
D. 110

14 **The magnitude and direction of electric field is _____**



A. 0
B. $60\ \frac{V}{m}$ upward
C. $60\ \frac{V}{m}$ downward
D. $600\ \frac{V}{m}$ downward

15 **The graph shows the relationship between pressure (P) and volume (V) for different thermodynamic processes. Which curve represents an adiabatic process?**

A. 1
B. 2
C. 3
D. 4

16 **Thermometer used to measure very low temperature is _____**

A. Thermocouple
B. Pyrometer
C. Gas thermometer
D. Magnetic thermometer

17 **If 1 gm of steam at 100°C is mixed with 1 gm of ice at 0°C, then resultant temperature of mixture will be _____**

A. 0°C
B. 5°C
C. 100°C
D. 50°C

18 **A long solenoid is formed by winding 20 turns per cm. What current is required to produce a magnetic field of 20 mT inside the solenoid?**

A. 8A
B. 0.8A
C. 80A
D. 4A

19 **Hubble's law is based on _____**

A. Wien's displacement law
B. Doppler effect
C. Cauchy's law
D. Stefan's law

20 **Which of the following pair of physical quantities have identical dimensions?**

A. Impulse and linear momentum
B. Frequency and wavelength
C. Pressure and energy
D. Impulse and force

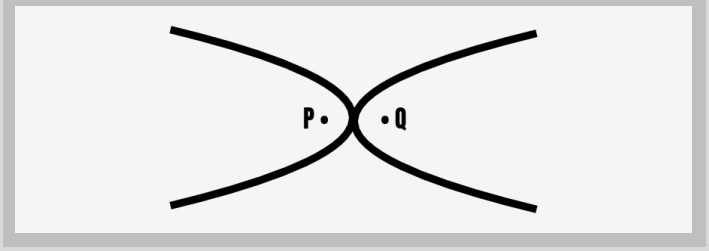
21 **NAND gate will produce a low output when _____**

A. All inputs are high
B. All inputs are low
C. Only one input is high
D. Only one input is low

22 **During α-emission, which of the following statements about the kinetic energy is true?**

A. The kinetic energy of the daughter nucleus is less than that of the α-particle
B. The kinetic energy of the daughter nucleus is more than that of the α-particle
C. The kinetic energy of the α-particle is equal to that of the daughter nucleus
D. Kinetic energy is not conserved

23 **The phase difference of the particles is _____**



A. 0
B. π^c
C. $\frac{\pi^c}{2}$
D. $2\pi^c$

24 **When a particle-antiparticle pair is annihilated, the energy liberated is _____**

A. $h\nu$
B. $2h\nu$
C. $nh\nu$
D. $(n+1)h\nu$

25 **A galvanometer can be converted into ammeter by connecting _____**

A. Low resistance in parallel
B. Low resistance in series
C. High resistance in parallel
D. High resistance in series

26 **A proton when accelerated through a potential difference of V volt has a wavelength λ associated with it. An α-particle in order to have same wavelength must be accelerated through a p.d. of _____**

A. V volt
B. $\frac{V}{2}$ volt
C. $\frac{V}{8}$ volt
D. 2V volt

27 **A student measures the acceleration due to gravity at a location as $6.65\ \frac{m}{s^2}$, $6.66\ \frac{m}{s^2}$, and $6.67\ \frac{m}{s^2}$. How would you classify his measurements?**

A. Accurate and precise
B. Precise but not accurate
C. Accurate but not precise
D. Neither accurate nor precise

28 **If the pressure amplitude of a sound wave is doubled, by how much does the intensity of the sound increase?**

A. 2 times
B. 3 times
C. $\sqrt{2}$ times
D. 4 times

29 A ship of mass 3×10^8 kg, initially at rest, is pulled by a force of 5×10^5 N and covers a distance of 3 m. Assuming negligible water resistance, what is the speed of the ship?

A. $0.5 \frac{m}{s}$ B. $0.1 \frac{m}{s}$
C. $0.2 \frac{m}{s}$ D. $1 \frac{m}{s}$

30 From a supply of identical capacitors, each rated 8 μF, 250 V, what is the minimum number of capacitors required to make a composite capacitor of 16 μF, 1000 V?

A. 32 B. 8
C. 1 D. 2

31 A simple pendulum has an amplitude R and time period T. The time taken by it to travel from $y = R$ to $y = \frac{R}{2}$ is _____

A. $\frac{T}{12}$ B. $\frac{T}{6}$
C. $\frac{T}{4}$ D. $\frac{T}{3}$

32 The force that binds proton and neutron in a nucleus is _____

A. Gravitational force
B. Weak nuclear force
C. Electromagnetic force
D. Strong nuclear force

33 The rate of flow of heat through a metal bar of cross sectional area 1 m^2 when the temperature gradient is $1^\circ \frac{C}{m}$ under steady state is called _____

A. Thermal conductivity
B. Specific heat capacity
C. Thermal resistance
D. Stefan's constant

34 An infinitely long straight wire produces a magnetic field of 3×10^{-5} T at a distance of 10cm from the wire. What is the current in the wire?

A. 15 A B. 10 A
C. 5 A D. 1 A

35 An electron moves with speed $10^5 \frac{m}{s}$ in an electric field $E = 2 \times 10^5 \frac{N}{C}$. What magnetic field must be applied perpendicular to the electron's motion direction so that it passes undeflected through the fields?

A. 0.3T B. 2T
C. 3T D. 22T

36 A lens is immersed in a liquid having refractive index same as the lens. The new focal length of the lens when immersed in liquid will be _____

A. Zero B. Infinite
C. Increase D. Decrease

37 Which of the following should be increased to increase the number of electrons emitted in photoelectric effect?

A. Intensity of incident radiation
B. Frequency of incident radiation
C. Stopping potential
D. Wavelength of incident radiation

38 A converging lens of focal length 5.0 cm is placed in contact with a diverging lens of focal length 10.0 cm, then the power of combination is _____

A. +10 D B. -10 D
C. +20 D D. -20 D

39 Barrier potential in PN junction diode is formed due to _____

A. Mobility of impurity
B. Applied external voltage
C. Diffusion of charge carrier
D. Drift of charge carrier

40 The moment of inertia of a body is I and coefficient of linear expansion is α. If temperature is increased, the new moment of inertia will be _____

A. >I B. <I
C. =0 D. 0

41 At what depth, from earth surface, value of acceleration due to gravity becomes $\frac{g}{4}$? (Radius of earth = R, acceleration due to gravity at surface = g)

A. $\frac{3R}{4}$ B. $\frac{R}{2}$
C. $\frac{R}{3}$ D. $\frac{2R}{3}$

42 A house is connected to a 220V supply protected by a 9A fuse. What is the maximum number of 60W bulbs that can be turned on in parallel without blowing the fuse?

A. 11 B. 22
C. 33 D. 44

43 A wave in a resonance tube is _____

A. Longitudinal and stationary
B. Longitudinal and progressive
C. Transverse and stationary
D. Transverse and progressive

44 The heat equation $\Delta U = \Delta Q + \Delta W$ represents _____

A. Heat is given to system and work is done on system
B. Heat is released by system and work is done by system
C. Heat is released by system and work is done on system
D. Heat is given to system and work is done by system

45 In Young's double slit experiment, two interfering waves have amplitudes 3 and 5 units respectively. What is the ratio of maximum intensity to minimum intensity?

A. $\frac{4}{1}$ B. $\frac{1}{4}$
C. $\frac{9}{16}$ D. $\frac{16}{1}$

46 During sunset, the Sun appears red because _____

A. Red light scatters the most
B. Red light scatters the least
C. Scattering of light is directly proportional to wavelength
D. Dispersion of light

47 First law of thermodynamics is based on _____

A. Conservation of entropy
B. Conservation of energy
C. Conservation of charge
D. Conservation of enthalpy

48 When electron transit from first excited state and second excited state to the ground state respectively, then the ratio of wavelength emitted during the two transition is _____

A. $\frac{32}{27}$ B. $\frac{27}{32}$
C. $\frac{4}{9}$ D. $\frac{9}{4}$

49 A metal block is stretched along its length, causing its cross-sectional area to reduce to $\frac{A}{2}$. What happens to its Young's modulus and modulus of rigidity?

A. Modulus of rigidity and Young's modulus increase
B. Modulus of rigidity decrease, Young's modulus increase
C. Young's modulus decrease, modulus of rigidity increase
D. Remains unchanged

50 In a metal block, eddy current is due to _____

A. Varying magnetic field
B. Steady magnetic field
C. Current passing through the circular coil
D. Potential difference applied across the metal block

51 The oxidation number of chlorine in sodium chlorate is _____

A. +1 B. +3
C. +5 D. +7

52 The order of reaction, $r = K[A]^{\frac{1}{2}}[B]$ is _____

A. 1 B. 2
C. 1.5 D. 2.5

53 In a chemical bond, when one atom attracts the shared electron pair more strongly towards itself, this property is known as _____

A. Hydrogen bonding B. Ionization energy
C. Electronegativity D. Electron affinity

54 Number of orbitals in 4th energy level is _____

A. 2 B. 8
C. 16 D. 32

55 % of nitrogen in urea is _____

A. 47 B. 18
C. 58 D. 32

56 Functional group present in glucose is _____

A. Aldehyde only
B. Ketone only
C. Aldehyde and alcohol both
D. Aldehyde and ketone both

57 Minimata disease is caused by _____

A. Hg B. Cd
C. Ag D. Zn

58 Dead burnt plaster is _____

A. $CaSO_4$ B. $CaSO_4 \cdot \frac{1}{2}H_2O$
C. $CaSO_4 \cdot 2H_2O$ D. CaO

59 The electrochemical cell in which hydrogen and oxygen react directly to produce electrical energy is called _____

A. Daniel cell B. Lead storage cell
C. Fuel cell D. Dry cell

60 Which of the following compounds yields hydrogen peroxide (H_2O_2) on treatment with dil. HCl?

A. PbO_2 B. Na_2O_2
C. MnO_2 D. TiO_2

61 Methyl orange indicator gives a red colour in the presence of _____

A. NaOH B. HCl
C. Na_2CO_3 D. NaCl

62 Which of the following compound have lowest octane number?

A. Iso-octane B. n-heptane
C. Toluene D. n-hexane

63 **Nascent hydrogen is _____**

A. Hydrogen gas absorbed on metal surface
B. Atomic hydrogen with excess energy
C. Molecular hydrogen
D. Isotope of hydrogen

64 **Seaweed (kelps) is the main source of _____**

A. Cl_2 B. I_2
C. Br_2 D. F_2

65 **Which type of lamp emits yellow light commonly used for highway and street lighting?**

A. Sodium-vapor lamp B. Argon lamp
C. Neon lamp D. Nitrogen lamp

66 **Molecular phosphorous (P_4) reacts with hot and conc. H_2SO_4 to give _____**

A. Phosphorous
B. Ortho-phosphoric acid
C. Meta-phosphoric acid
D. Pyro-phosphoric acid

67 **Hydrolysis of isocyanides gives _____**

A. Primary amine and CO_2
B. Primary amine and carboxylic acid
C. Primary amine and secondary amine
D. Secondary amine and CO_2

68 **Which of the following tests can be used to distinguish primary (1°), secondary (2°), and tertiary (3°) amines?**

A. Hoffmann's method B. Nitrous acid test
C. Silver mirror test D. Dil. NaOH

69 **Nitrogen gas is chemically inert due to _____**

A. Low electronegativity
B. Absence of valance electron
C. Presence of triple bond between nitrogen atoms
D. Lack of H-bond

70 **Which of the following statement about Hydrogen is correct?**

A. Hydrogen has same IP as alkali metals
B. It has oxidation number of +1 and -1
C. It will not be liberated at anode
D. It has same EN as halogen

71 **H_2S is a group reagent for _____**

A. Group I B. Group II
C. Group III D. Group IV

72 **12 g of magnesium (Mg) reacts with 12 g of oxygen (O_2) to form magnesium oxide (MgO). The residue contains _____**

A. 20 gm of MgO and 4 gm of O_2
B. 24 gm of MgO only
C. 20 gm of MgO and 6 gm of O_2
D. 16 gm of MgO and 8 gm of O_2

73 **The solubility of magnesium hydroxide, $Mg(OH)_2$ is minimum in _____**

A. Pure water B. NaOH
C. HCl D. NH_4Cl

74 **When zinc (Zn) reacts with alkaline sodium nitrate ($NaNO_3$), which of the following is formed?**

A. NH_3 B. NO_2
C. N_2 D. NO

75 **When FeS reacts with dilute H_2SO_4, H_2S gas is evolved, giving a visible black colour in which of the following tests?**

A. Lead acetate test B. $CuCl_2$ test
C. $KMnO_4$ test D. $PbCl_2$ test

76 **The solubility of alkaline earth metal hydroxides is less than that of alkali metals hydroxide because _____**

A. They have higher lattice energy
B. They have lower lattice energy
C. They have higher hydration energy
D. They have lower melting point

77 **When benzene is treated with concentrated H_2SO_4 and concentrated HNO_3 at 100°C, which of the following compounds is formed?**

A. Nitrobenzene B. p-dinitrobenzene
C. m-dinitrobenzene D. o-dinitrobenzene

78 **The ratio of rate of diffusion of SO_2 and O_2 is _____**

A. $\frac{\sqrt{2}}{1}$ B. $\frac{1}{\sqrt{2}}$
C. $\frac{2}{1}$ D. $\frac{1}{2}$

79 **If $KMnO_4$ is treated with 0.5 L of oxalic acid of 10^{-2} M and 10^{-4} M respectively, the amount of $KMnO_4$ required will be _____**

A. More for 10^{-4} M oxalic acid
B. More for 10^{-2} M oxalic acid
C. Equal volume for both concentration of oxalic acid
D. Insufficient data

80 When ammonia (NH_3) is treated with excess ethyl chloride (C_2H_5Cl), the product formed is _____

A. Ethanamine
B. N-ethyl ethanamine
C. N, N-diethylethanamine
D. Tetraethyl ammonium chloride

81 Ethene and Ethyne can be distinguished by _____

A. Conc. H_2SO_4
B. Alkaline $KMnO_4$
C. Br_2 in CCl_4
D. $AgNO_3$ in NH_4OH

82 Which of the following statements correctly describes a homologous series?

A. All members have identical physical property
B. Successive members differ by 14 amu in molecular mass
C. The first member resembles all other members in physical properties
D. Successive member differ by $-CH_3$ group

83 Benzene exists in liquid state due to _____

A. Vander Waal force
B. Dipole-dipole interaction
C. Hydrogen bonding
D. London-dispersive force

84 Haloalkane are more reactive than hydrocarbon due to _____

A. Formation of carbonation
B. Polar bond between C and X
C. Absence of H-bond
D. Presence of halogen

85 Bayer's reagent is _____

A. Acidified $K_2Cr_2O_7$
B. Acidified $KMnO_4$
C. Alkaline $KMnO_4$
D. Neutral $FeCl_3$

86 Sodium extract is used _____

A. to detect functional group
B. as an oxidizing agent
C. as a reducing agent
D. for detection of foreign element in organic compound

87 Ozone is prepared by passing oxygen through _____

A. UV light
B. Electric arch
C. Moist Ag_2O
D. Red hot platinum

88 In esterification, alcohol and carboxylic acid reacts in presence of _____

A. conc. NaOH
B. conc. H_2SO_4
C. Na_2CO_3
D. $K_2Cr_2O_7$

89 Which of the following hydrocarbon is commonly used for cutting and welding purpose?

A. Propyne
B. Butyne
C. Acetylene
D. Methane

90 When sugar reacts with concentrated H_2SO_4, it turns black. This is due to _____

A. Oxidation
B. Reduction
C. Dehydration
D. Polymerization

91 In Friedel craft acylation, the function of $AlCl_3$ is _____

A. act as catalyst
B. act as lewis acid and generate acylium ion
C. act as lewis base
D. generate nucleophile

92 Which of the following best represents an example of an addition polymer?

A. Polyethylene from ethylene
B. Bakelite from phenol and formaldehyde
C. Nylon-6,6 from hexamethylenediamine and adipic acid
D. Polyester from ethylene glycol and terephthalic acid

93 In a simple cubic unit cell, how many atoms are contributed by each corner to the unit cell?

A. 1
B. $\frac{1}{2}$
C. $\frac{1}{4}$
D. $\frac{1}{8}$

94 In Victor Meyer's test, red colour is given by _____

A. Primary alcohol
B. Secondary alcohol
C. Tertiary alcohol
D. Phenol

95 ZnO is amphoteric because _____

A. It reacts with acid to form salt and water
B. It reacts with base to form salt and water
C. It reacts with both acid and base
D. It does not react with either acid or base

96 Oleic acid is _____

A. Unsaturated fatty acid
B. Saturated fatty acid
C. Essential amino acid
D. Polysaccharide

97 Which is not a property of liquid?

A. Intermolecular force is negligible
B. Surface tension decrease with increase in temperature
C. At high elevation, BP decreases
D. Vapor pressure increases with temperature

98 Which of the following reactions is favored by high temperature and high pressure?

A. $N_2 + 3H_2 + 22.08\ Kcal \rightarrow 2NH_3$
B. $2Cl_2O_7 \rightarrow 2Cl_2 + 7O_2 + 126.8\ Kcal$
C. $Cl_2 + O_7 \rightarrow 2ClO_2 - 49.4\ Kcal$
D. $2NF_3 \rightarrow N_2 + 3F_2 + 54.4\ Kcal$

99 **Which orbital is filled after filling 3d-orbitals?**

A. 4s
B. 4p
C. 3f
D. 4d

100 **The order of reaction can be determined by _____**

A. Counting the number of reactants in elementary reaction
B. By experimenting multiple times
C. Looking at the reacting vessel
D. Counting total number of molecules in all steps

101 **Which of the following gastric cells indirectly help in erythropoiesis?**

A. Chief cells
B. Mucous cells
C. Goblet cells
D. Parietal cells

102 **A triploblastic animal with bilaterally symmetrical larva and asymmetrical adult is _____**

A. Arachnida
B. Bivalvia
C. Gastropoda
D. Echinodermata

103 **A marine cartilaginous fish that can produce electric current is _____**

A. Pristis
B. Torpedo
C. Rheo
D. Scoliodon

104 **Tetraneury (tetra-neuron) is a characteristic feature of the nervous system of which of the following animal groups?**

A. Annelida
B. Mollusca
C. Coelenterata
D. Platyhelminthes

105 **Find the correctly matched pair _____**

A. Porifera: Sea pen
B. Platyhelminthes: Nereis
C. Echinodermata: Leech
D. Mastigophora: Volvox

106 **Which type of cell is found in the gemmules of sponges?**

A. Archeocytes
B. Myocytes
C. Trophocytes
D. Scleroblast

107 **Myelin sheath provides insulation to CNS, which of the following glial cells forms insulation sheath in CNS?**

A. Schwann cells
B. Astrocytes
C. Dendrites
D. Oligodendrocytes

108 **Totipotent cells present in Platyhelminthes is _____**

A. Interstitial cells
B. Archeocytes
C. Neoblasts
D. Absent

109 **Which of the following is the diagnostic confirmatory test for tuberculosis (TB)?**

A. Sputum test
B. Mantoux test
C. Chest X-ray
D. Bronchoscopy

110 **In humans, the testes are located outside the abdominal cavity primarily to _____**

A. Facilitate easier fertilization
B. Maintain scrotal temperature, 2–3°C lower than the body temperature
C. Accelerate maturation of sperm
D. Protect them from injury

111 **Lysolecithin produced during Plasmodium infection is secreted by _____**

A. Spleen
B. Liver
C. RBCs
D. Kidney

112 **Radial symmetry and ambulacral ring vessel is found in _____**

A. Asterias
B. Unio
C. Helix
D. Limpet

113 **The ends of long bones in the limbs are covered by which type of cartilage?**

A. Hyaline cartilage
B. Elastic cartilage
C. Spongy cartilage
D. Calcified cartilage

114 **Sudden lapse of memory after an emotional event maybe due to _____**

A. Cardiogenic shock
B. Neurogenic shock
C. Septicemia
D. Stroke

115 **Sexual dimorphism in frogs is primarily due to the presence of which feature in males?**

A. Vocal sacs
B. Vocal cord
C. Cloacal aperture
D. Nuptial pads on the forelimbs

116 **Which of the following substance, if introduced into blood stream, would cause coagulation of blood at the site of its introduction?**

A. Thromboplastin
B. Heparin
C. Fibrinogen
D. Prothrombin

117 **Injury to vagus nerve is less likely to affect _____**

A. Tongue movement
B. Pancreatic secretion
C. Cardiac movement
D. Gastrointestinal movement

118 **After an accident a person loses his ability to smell, it might be due to damage in _____**

A. Optic chiasma
B. Olfactory bulb
C. Cerebellum
D. Mid brain

119 **Kappa-particles are associated with transfer of genetic information through _____**

A. Sex-linked inheritance
B. Cytoplasmic inheritance
C. Macro inheritance
D. Nuclear inheritance

120 **Blockage of coronary artery may likely cause _____**

A. Heart attack
B. Stroke
C. Hemorrhage
D. Rheumatoid arthritis

121 **A young infant may be feeding entirely on mother's milk which is white in colour but the stool which the infant passes out is quite yellowish. The yellow colour of stool due to _____**

A. Pancreatic juice poured into duodenum
B. Intestinal juice
C. Undigested milk protein casein
D. Bile pigments passed through bile juice

122 **Pecten, a comb like structure is present in the eyes of _____**

A. Mammals
B. Birds
C. Snakes
D. Pisces

123 **In a copulating pair of earthworm, which two processes take place?**

A. Reciprocal and external fertilization
B. Cross and reciprocal fertilization
C. Cross and internal fertilization
D. Reciprocal and internal fertilization

124 **When a transport maximum and renal threshold of a substance is reached, what happens to the re-absorption of substance?**

A. Increase
B. Decrease
C. Remains same
D. Either increase or decrease

125 **Two different species cannot live for long duration in the same niche or habitat. This is _____**

A. Allen's law
B. Gause's hypothesis
C. Dollo's rule
D. Weismann's theory

126 **Which of the following is formed as end product of erythrocytic schizogony in human RBCs?**

A. Sporozoite
B. Trophozoite
C. Merozoites
D. Cryptomerozoites

127 **Chloragogen cell present in the coelomic fluid of earthworm are analogous to _____**

A. Vertebrate liver
B. Vertebrate kidney
C. Coelenterates cnidoblast
D. Flame cells

128 **Which of the following structures help in equilibrium and body balance in humans?**

A. Basilar membrane and macula
B. Basilar membrane and otolith
C. Organ of corti and tectorial membrane
D. Cristae ampullaris and macula

129 **Low pH of gastric juice, lysozyme in tears and saliva, and high body temperature prevent the growth of pathogens. These mechanisms represent which type of immunity?**

A. Innate anatomical barrier
B. Innate physiological barrier
C. Natural passive immunity
D. Active immunity

130 **Which of the following pairs represents homologous organs?**

A. Wings of bat and wings of cockroach
B. Pectoral fins of fish and forelimbs of horse
C. Air sacs of fish and lungs of frog
D. Wings of bird and wings of insect

131 **The periodic reversal of blood flow due to peristalsis is seen in _____**

A. Herdmania
B. Amphioxus
C. Asterias
D. Chordata

132 **Before 12 weeks of pregnancy, during antenatal screening, amniocentesis is usually not suggested because _____**

A. Risk of harm to fetus before 12 weeks
B. Can't detect genetic disorder
C. Can't detect neural tube defect at this stage
D. Need to give general anesthesia

133 **Which hormone is commonly referred to as the "life-saving" hormone?**

A. Adrenaline
B. Aldosterone
C. Calcitonin
D. Thyroxine

134 **Universal blood donor group is _____**

A. O^-
B. O^+
C. AB^-
D. AB^+

135 **Which stage of plasmodium is likely to be found in the stomach of female Anopheles mosquito, which has just taken a blood meal from malaria patient?**

A. Merozoites, trophozoites, gametocytes
B. Sporozoites, trophozoites, and gametes
C. Merozoites, gametocyte and zygote
D. Gametocytes only

136 **A frog species living in a terrestrial niche with limited access to water has evolved shorter hind limbs. This morphological change represents which trade-off?**

A. Increase swimming efficiency
B. Decrease jumping ability for camouflage advantage
C. Increase cellulose digestion
D. Absence of pond in forest

137 **Deuterostome, enterocoelomate organisms with mesodermal calcareous endoskeleton is _____**

A. Ascidia
B. Sepia
C. Pila
D. Asterias

138 **Which of the following tissues has the least regenerative capacity?**

A. Muscles
B. Eye lens
C. Iris of eye
D. Neuron

139 **Primary function of sodium-potassium pump is _____**

A. Pump K^+ out and Na^+ into cell
B. Pump K^+ in and Na^+ out
C. Pumps both out
D. Pumps both in

140 **Homo erectus (1.25 MYA) is similar to modern man such that limb proportion is similar to us and they make the use of tools, which of the following is significance of this?**

A. Essentially ate fruits
B. Travelled and haunted animals
C. Did not use fire
D. Stay limited to forest

141 **The correct information flow in eukaryotic cell involves RNA processing. Choose the right sequence _____**

A. DNA $\rightarrow$ mRNA $\rightarrow$ Protein
B. DNA $\rightarrow$ hnRNA $\rightarrow$ mRNA $\rightarrow$ Protein
C. DNA $\rightarrow$ Protein $\rightarrow$ mRNA
D. DNA $\rightarrow$ rRNA $\rightarrow$ mRNA $\rightarrow$ Protein

142 **Which of the following have same genotype and phenotype ratio?**

A. Incomplete dominance
B. Complete dominance
C. Dihybrid cross
D. Multiple allelism

143 **Which of the following bio fertilizers is effectively used in paddy (rice) fields?**

A. Rhizobium
B. Azolla
C. Marchantia
D. Fern

144 **Embryo sac in angiosperms develops from _____**

A. Microspore
B. Functional megaspore
C. Tapetum
D. Nucellus

145 **Cork cells are impermeable in water due to deposition of waxy substance called _____**

A. Lignin
B. Suberin
C. Pectin
D. Cellulose

146 **Which of the following cell is living and metabolically active?**

A. Tracheids
B. Vessel element
C. Companion cell
D.

147 **Phylloclade and cladode are morphological adaptation for photosynthesis and water storage, found in _____**

A. Hydrophytes
B. Mesophytes
C. Xerophytes
D. Halophytes

148 **How many different phenotypes are possible in the human ABO blood group system?**

A. 2
B. 3
C. 4
D. 5

149 **Bryophytes differ from Pteridophytes in having _____**

A. Vascular bundles
B. Dependent sporophyte
C. Seed habit
D. Independent gametophyte

150 **Which of the following is common between eukaryotes and prokaryotes?**

A. Nucleus
B. Mitochondria
C. Histone protein
D. Genetic code

151 **Glycolysis takes place in _____**

A. Cytoplasm
B. Mitochondria
C. Chloroplast
D. Ribosome

152 **During investigation, a person's blood contains both A & B protein. In this case, AB blood group in human is an example of _____**

A. Incomplete dominance
B. Co-dominance
C. Epistasis
D. Complete-dominance

153 **Four chromatids and two centromere of homologous chromosome is seen in _____**

A. Pachytene
B. Zygotene
C. Leptotene
D. Diplotene

154 **Deepest algae found in deep sea is _____**

A. Red algae
B. Brown algae
C. Green algae
D. Blue-green algae

155 **Which of the following cell organelle is not bound by membrane?**

A. Nucleus
B. Plastid
C. Golgi body
D. Ribosome

156 **Decomposition of organic nitrogen of dead animals and plant into ammonia is _____**

A. Nitrogen fixation
B. Ammonification
C. Nitrification
D. Denitrification

157 **When two genes are very close to each other and tends to inherit together. This phenomenon is known as _____**

A. Epistasis
B. Crossing over
C. Complete linkage
D. Incomplete linkage

158 **Which of the following phenomenon leads to the corrosion of metal structure like bridges and statues?**

A. Acid rain
B. Green house effect
C. Bio-magnification
D. Eutrophication

159 **Brinjal, Potato, Tomato, Onion, Ginger belong to _____**

A. A single family
B. Four genera
C. Five genera
D. Same genera

160 **Bicollateral vascular bundle is present in _____**

A. Cucurbits
B. Monocot roots
C. Fern
D. Yucca

161 **The first stable compound of Calvin cycle is _____**

A. PEP
B. PGA
C. PGAL
D. Glucose

162 **In which type of inflorescence does the main axis show definite growth and terminate in a flower (bud at the tip)?**

A. Racemose
B. Cymose
C. Verticillaster
D. Hypanthidium

163 **Tapetum is found in _____**

A. Anther
B. Female gametophyte
C. Microspore
D. Male gametophyte

164 **A sudden, heritable change in the genetic material that introduces variation is called _____**

A. Mutation
B. Polyploidy
C. Genetic variation
D. Linkage

165 **Which of the following does not occupy any trophic level?**

A. Producer
B. Food web
C. Consumer
D. Decomposers

166 **Which plant hormone delays senescence and counteracts apical dominance?**

A. Auxin
B. Cytokinin
C. Gibberellin
D. ABA

167 **When a tetraploid male plant is crossed with tetraploid female plant. The endosperm will be _____**

A. Tetraploid
B. Octoploid
C. Hexaploid
D. Triploid

168 **If a botanist found that a plant reproduces by spore not by seed, then on which group it will be likely to be placed?**

A. Bryophytes
B. Sporophyte
C. Cryptogams
D. Thallophyte

169 **If two pea plants having red coloured flower with unknown genotype are crossed, 75% of flowers are red and 25% white. The genotype constitution of the two parents would be _____**

A. Both heterozygous
B. Both homozygous
C. Both hemizygous
D. One homozygous, another heterozygous

170 Grass $\rightarrow$ Grasshopper $\rightarrow$ Frog $\rightarrow$ Snake $\rightarrow$ Eagle**. The type of food chain is _____**

A. Parasitic
B. Grazing
C. Predatory
D. Detritus

171 **Multiflagellated antherozoid is found in which group of plants?**

A. Moss
B. Fern
C. Algae
D. Fungi

172 **Grafting is not possible in monocot plants because of _____**

A. Open vascular bundle
B. Presence of scutellum
C. Absence of vascular cambium
D. Presence of scattered vascular bundles

173 **Smallest segment of DNA that undergoes mutation is _____**

A. Recon
B. Muton
C. Cistron
D. Gene

174 **Enzyme used in genetic engineering is _____**

A. Restriction endonuclease
B. DNA ligase
C. DNA polymerase
D. Endonuclease

175 **Spirogyra after conjugation forms zygote, the zygote then undergoes _____**

A. Meiosis
B. Mitosis to form new filament directly
C. Germination without division
D. Zygote remains same

176 Corolla with claw and blade (hand type) is found in _____

A. Solanaceae
B. Brassicaceae
C. Liliaceae
D. Fabaceae

177 Which of the following is not a characteristic of Liliaceae family?

A. Tetramerous flower
B. Trilocular ovary with axile placentation
C. Epiphyllous stamen
D. Usually herbs

178 Plasmogamy, karyogamy and meiosis in a sequence is a characteristics feature of _____

A. Fungi
B. Lichen
C. Bryophytes
D. Algae

179 The main characteristic of mitosis but not seen in meiosis is _____

A. Homologous chromosome behave independently
B. Chromatids separate in anaphase
C. Crossing over between homologous chromosome
D. Pairing of homologous chromosomes to form bivalents

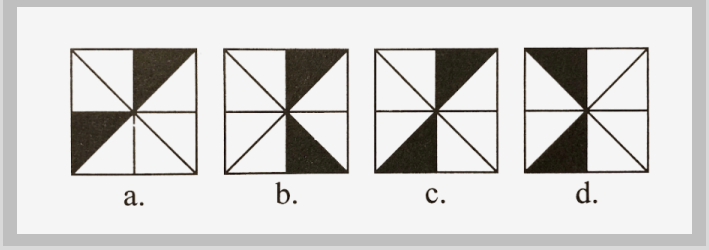
180 The cell in which conversion of glucose to CO_2 does not occur aerobically is _____

A. RBC
B. WBC
C. Tissue
D. Platelets

181 The letter 'L', is rotated 270° counter-clockwise, it is equivalent to _____

A. 90° clockwise
B. 90° counter-clockwise
C. 180° clockwise
D. 180° counter-clockwise

182 Choose the odd one out.

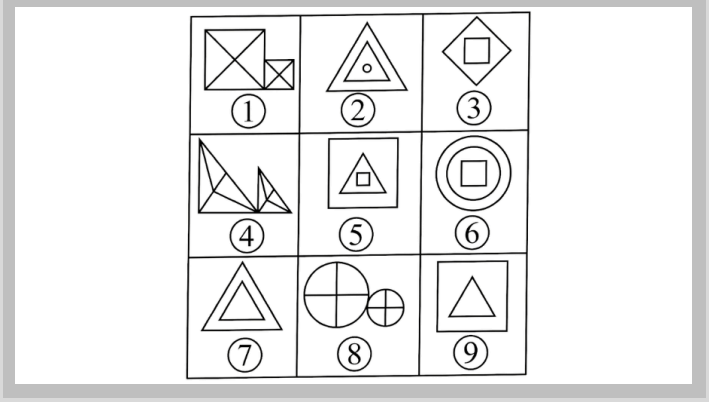


A. a
B. b
C. c
D. d

183 89, 89, 75, 75, 61, 61, ?

A. 53
B. 51
C. 47
D. 43

184 Group the given figures into three classes using each figure only once.



A. 1, 3, 7 ; 2, 4, 6 ; 5, 8, 9
B. 1, 4, 6 ; 2, 5, 7 ; 3, 8, 9
C. 1, 4, 8 ; 2, 5, 6 ; 3, 7, 9
D. 1, 4, 8 ; 2, 7, 9 ; 3, 5, 6

185 During sunrise a person face towards the poll and shadow of the poll falls to left of the person. In which direction is he facing now?

A. North
B. South
C. East
D. West

186 The average of 7 numbers is 14 and one of the numbers '20' is subtracted from total then the new average is _____

A. 5
B. 3
C. 14
D. 11

187 Find the term that should fit in the given series to complete following letter series?
$AN_{14}, BO_{17}, DQ_{21}, ER_{23}$

A. CI_{14}
B. CO_{15}
C. DP_{16}
D. CP_{19}

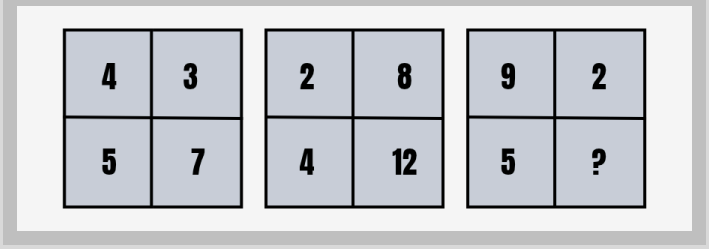
188 Three doctors and three nurses are sitting alternately on a round table, facing towards the centre. After a break, a doctor was replaced by a pharmacist. Again, after lunch, a doctor between the nurses was replaced by technician and the doctor to the right of the nurse was replaced by physiotherapist, then the person opposite to pharmacist is _____

A. Doctor
B. Nurse
C. Technician
D. Physiotherapist

189 The principal of 6 medical college are A, B, C, D, E and who are sitting around circular table facing towards centre B is between D and C. A is between E and C, then E is between A and F. Then who is to the immediate left of E?

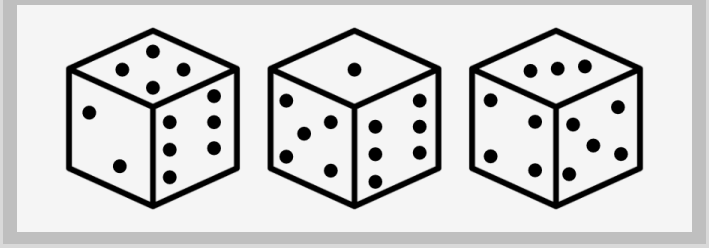
A. A
B. C
C. F
D. B

190 **Find the missing number.**



A. 11 B. 12
C. 13 D. 14

191 **How many dots lie in the opposite dot?**



A. 6 B. 2
C. 4 D. 3

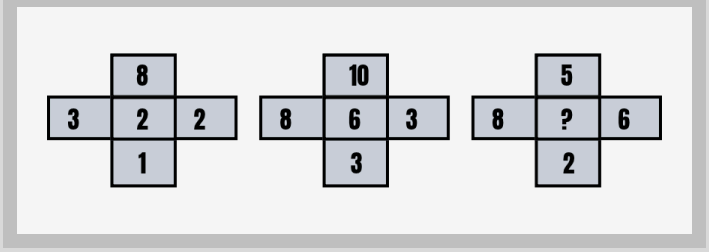
192 **Two trains, 100 m long and 80 m long are running with the speed of 51 $\frac{m}{s}$ and 42 $\frac{m}{s}$ respectively in the same direction. After what time, they both will cross each other?**

A. 20 sec B. 25 sec
C. 30 sec D. 35 sec

193 **Clock gains 24 minutes in 48 hours. Clock is set at 8 am on Thursday. What time will the clock show at 8 am on Monday?**

A. 8:24 am B. 8:48 am
C. 8:00 am D. 9:48 am

194 **Find the missing number from the figure**



A. 34 B. 36
C. 38 D. 40

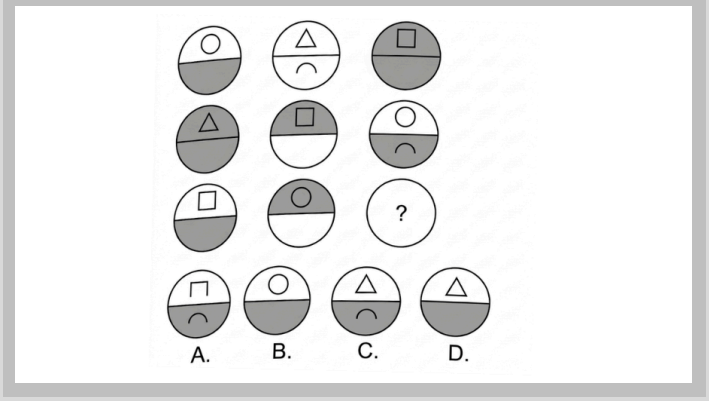
195 **A, B and C own a land. If A has one half more than B. B has one and half more than C. If the total land of A, B and C is 76 ropanis. Then land own by A, B and C respectively is _____**

A. 26, 30, 12 B. 36, 24, 16
C. 24, 36, 16 D. 37, 20, 14

196 **Ramesh is the eldest who is handling the transportation. Raj and Ram are of same age handling game. Ram is younger than Raju. Rajnish is elder brother of Raju who is handling the camera. Who is the second eldest brother among them?**

A. Ramesh B. Raj
C. Rajnish D. Ram

197 **The correct alternative in the missing place is**



A. A B. B
C. C D. D

198 **Diana is going to a ..... outside Henry town, and her friends live near the airport. Maria lives 10 miles from the airport, Alexis lives 5 miles away, and Henry lives 7 miles away from the airport, Diana is farther than Alexis but closer than Henry. What is Diana's distance from the airport?**

A. 4 miles B. 6 miles
C. 8 miles D. 11 miles

199 **Mr. Karki like panting landscape and walking while Mrs. Karki enjoys cooking and dancing. Both of them share a common interest in walking during the morning. Based on this information, which of the following statement about their interest is correct?**

A. Mrs. Karki does not like cooking
B. Mr. Karki likes painting and walking
C. Mr. Karki likes cooking
D. Mrs. Karki like painting

200 **A medical college has to conduct the exam of seven different subjects starting from 12th to 19th May: Microbiology, Physiology, Surgery, Anatomy, Biochemistry, Pathology and Pharmacology. Based on given information, which subject exam is on Wednesday? (i) The exam should start with Physiology. (ii) 13th of May was Sunday, being holiday. (iii) Exam should end with Biochemistry. (iv) The exam that conducted after holiday was surgery. (v) Anatomy was on the day before Biochemistry. (vi) Pathology was between Pharmacology & Microbiology.**

A. Surgery B. Pharmacology
C. Microbiology D. Pathology